FNCE 30007 DERIVATIVE SECURITIES

Tutorial Questions

Futures

Question 1

The part with a short position in a futures contract sometimes has options as to the precise asset that will be delivered, where delivery will take place, when delivery will take place, and so on. Do these options increase or decrease the futures price? Explain your reasoning.

Question 2

Explain how margins protect investors against the possibility of default.

Question 3

An investor enters into two long July futures contracts on frozen orange juice. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is $6,000 per contract, and the maintenance margin is $4,500 per contract. What price change would lead to a margin call? Under what circumstances could $2,000 be withdrawn from the margin account?

Question 4

Show that, if the futures price of a commodity is greater than the spot price during the delivery period, there is an arbitrage opportunity. Does an arbitrage opportunity exist if the futures price is less than the spot price? Explain your answer.

Question 5

On July 1, 2025, a U.S. company enters into a forward contract to buy 10 million British pounds on January 1, 2026. On September 1, 2025, it enters into a forward contract to sell 10 million British pounds on January 1, 2026. Describe the profit or loss the company will make in dollars as a function of the forward exchange rates on July 1, 2025, and September 1, 2025.

Question 6

What do you think would happen if an exchange started trading a contract in which the quality of the underlying asset was incompletely specified?

Question 7

An airline executive has argued: ‘There is no point in our using oil futures. There is just as much chance that the price of oil in the future will be less than the futures price as there is that it will be greater than this price.’ Discuss the executive’s viewpoint.